Link to code

Solution 1

1. $E[X] = \frac{1}{50}(3 \cdot 25 + 10 \cdot 15 + 5 \cdot 10) = \frac{11}{2}$
2. $E[X^2] = \frac{1}{50}(3^2 \cdot 25 + 10^2 \cdot 15 + 5^2 \cdot 10) = \frac{79}{2}$

   $(E[X])^2 = \frac{121}{4}$

   $\mathrm{Var}(X) = \frac{79}{2} - \frac{121}{4} = \frac{37}{4}$
3. $E[X] = \frac{1}{47}(3 \cdot 23 + 10 \cdot 15 + 5 \cdot 9) = \frac{264}{47}$
4. $E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X] = \mathrm{Var}(X) + (E[X])^2 - E[X] = 2 + 10^2 - 10 = 92$

Solution 2

Let F be the event that an email has "free money", and let S be the event that an email is spam.

1. $P(F) = P(F|S)P(S) + P(F|\neg S)P(\neg S)$

   $P(F) = \frac{1}{10} \cdot \frac{8}{10} + \frac{1}{100} \cdot \frac{2}{10}$

   $P(F) = \frac{41}{500}$
2. $P(S|F) = \frac{P(F|S)P(S)}{P(F|S)P(S)+P(F|\neg S)P(\neg S)}$

   $P(S|F) = \frac{\frac{1}{10}\cdot\frac{8}{10}}{\frac{1}{10}\cdot\frac{8}{10}+\frac{1}{100}\cdot\frac{2}{10}}$

   $P(S|F) = \frac{40}{41}$
3. $P(\neg S|\neg F) = \frac{P(\neg F|\neg S)P(\neg S)}{P(\neg F|\neg S)P(\neg S)+P(\neg F|S)P(S)}$

   $P(\neg S|\neg F) = \frac{\frac{2}{10}\cdot\frac{99}{100}}{\frac{2}{10}\cdot\frac{99}{100}+\frac{8}{10}\cdot\frac{9}{10}}$

   $P(\neg S|\neg F) = \frac{11}{51}$

Solution 3

1. Row-reduce D_1 and D_2:

$$D_1 = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 2 & 4 & -2 \\ -1 & 0 & -3 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$D_2 = \begin{pmatrix} 1 & -1 & 0 & 2 \\ -1 & 0 & 1 & -1 \\ 0 & -2 & 0 & 1 \\ 3 & 0 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

   So the columns of D_1 are linearly dependent, and the columns of D_2 are linearly independent.

2. The maximum number of columns that are linearly independent is 3 for D_1 and 4 for D_2.

3. The rank is 3 for D_1 and 4 for D_2, using the RREF's found in (1)

4. The sizes only work for $D_1\theta$ and $\theta^T D_1$:

$$D_1\theta = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 2 & 4 & -2 \\ -1 & 0 & -3 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

$$\theta^T D_1 = \begin{pmatrix} -1 & 0 & 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 2 & 4 & -2 \\ -1 & 0 & -3 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} -2 & -2 & -7 & 1 \end{pmatrix}$$

Solution 4

Let Anton = 1, Geraldine = 2, James = 3, Lauren = 4.
Let English = 1, French = 2, German = 3, Italian = 4, Spanish = 5.

1. $A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix}$

   $$AA^T = \begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 2 & 2 \\ 0 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

   The (i, j)-th entry in AA^T is the number of shared languages between person i and person j. When $i = j$, the (i, j)-th entry counts the number of languages person i speaks.

   $$A^T A = \begin{pmatrix} 3 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 1 & 1 \\ 3 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 & 2 \end{pmatrix}$$

   The (i, j)-th entry in $A^T A$ is the number of shared people between language i and language j. When $i = j$, the (i, j)-th entry counts the number of people who speak language i.

2. $A = \begin{pmatrix} -8 & -8 & 9 \\ 1 & 5 & -6 \\ -5 & -10 & -2 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -0.159 & -0.241 & 0.007 \\ 0.073 & 0.139 & -0.089 \\ 0.034 & -0.091 & -0.073 \end{pmatrix}$

   $$B = \begin{pmatrix} 3 & -2 & -2 \\ -2 & -5 & -4 \\ -3 & -5 & -1 \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 0.273 & -0.145 & 0.036 \\ -0.182 & 0.164 & -0.291 \\ 0.091 & -0.382 & 0.345 \end{pmatrix}$$

$$C = \begin{pmatrix} -2 & -9 & 10 \\ -7 & 0 & -5 \\ -9 & -10 & 0 \end{pmatrix}, \quad C^{-1} = \begin{pmatrix} -0.127 & -0.253 & 0.114 \\ 0.114 & 0.228 & -0.203 \\ 0.177 & 0.154 & -0.159 \end{pmatrix}$$

Solution 5

1. Lowest mean: `long` (-122.214), higest mean: `price` (540,088.142).
   Lowest variance: `waterfront` (0.007), higest variance: `price` (134,776,142,225.573).

2. All of the features are positively correlated with `price`:

feature	corr with `price`
`long`	0.021626
`condition`	0.036362
`yr_built`	0.054012
`sqft_lot15`	0.082447
`sqft_lot`	0.089661
`yr_renovated`	0.126434
`floors`	0.256794
`waterfront`	0.266369
`lat`	0.307003
`bedrooms`	0.308350
`sqft_basement`	0.323816
`view`	0.397293
`bathrooms`	0.525138
`sqft_living15`	0.585379
`sqft_above`	0.605567
`grade`	0.667434
`sqft_living`	0.702035

   The feature with the highest correlation is `sqft_living`.

3. No